

SEPM Assignment 1

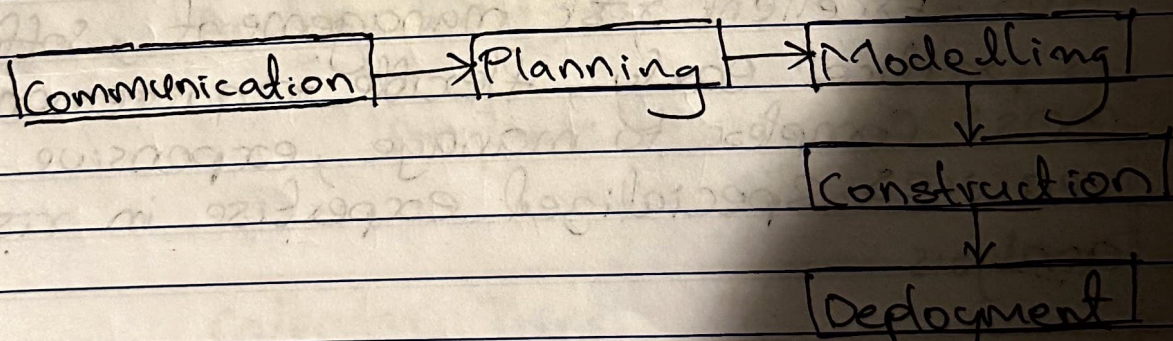
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Q Explain different Software Process Models.

ANS Software Process Models (also called SDLC models) are various strategies or frameworks used to plan, manage and execute the development of a software product.

1. Waterfall Model

The waterfall model is the oldest and most traditional approach. It is a linear-sequential model where the development flows steadily downwards (like a waterfall) through clearly defined phases. Each phase must be completed before the next one begins.



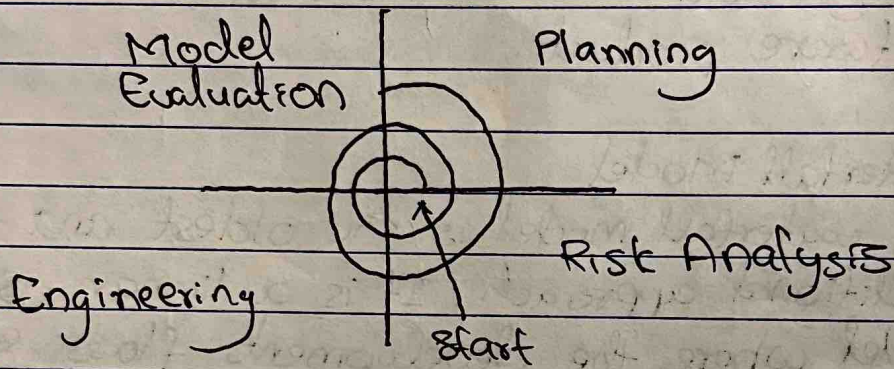
Best for projects with fixed, stable requirements where changes are unlikely (e.g., banking systems, government projects)

Pros: Simpler to manage, clear milestones, requirements are well understood.

Cons: Extremely rigid, difficult to go back and change things.

2. Spiral Model

The Spiral model combines the relative model of Agile with the controlled, systematic aspects of waterfall. Its primary focus is Risk Analysis. The project passes through four phases repeatedly in a spiral trajectory.



Best for large, high-risk or complex projects (e.g. new product lines, research and development)

Pros: Excellent risk management, software is produced early, flexible requirements
Cons: Complex to manage, expensive, requires specialized expertise in risk analysis.

3. Incremental Model

In the Incremental model, requirements are broken down into multiple standalone modules. The software is developed and delivered in "increments". The first increment provides the core functionality and subsequent increments add more features.

Phases in each increment: ① Requirement Analysis ② Design ③ Implementation ④ Testing ⑤ Deployment

Increment 1 → Core features

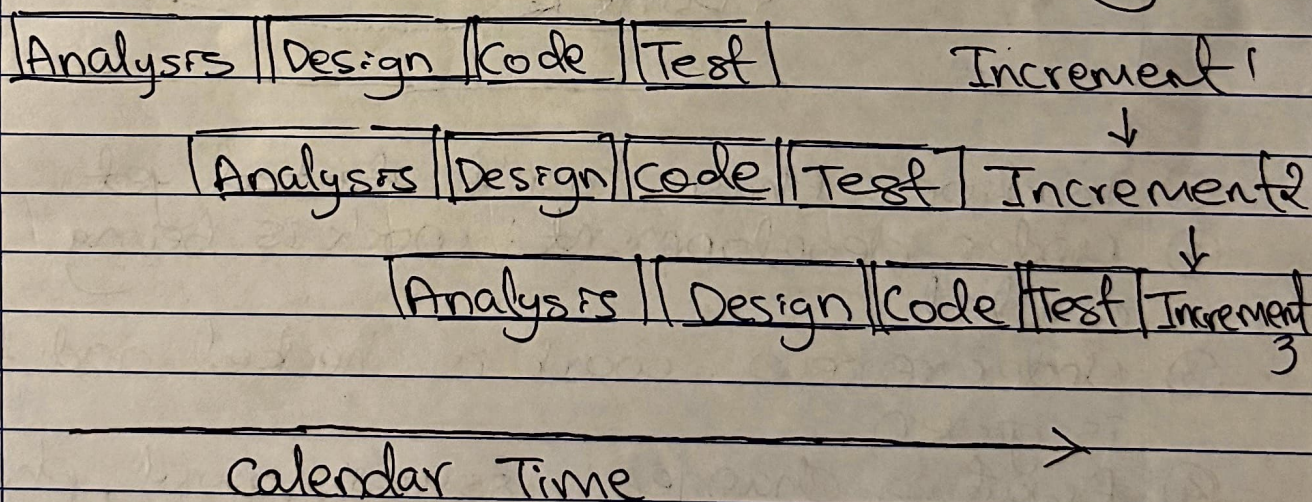
Increment 2 → Additional features

Increment 3 → Advanced features

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Increment n → Final System

In this software is delivered in parts, Each release is functional, Early working delivery of working software, user feedback is incorporated continuously

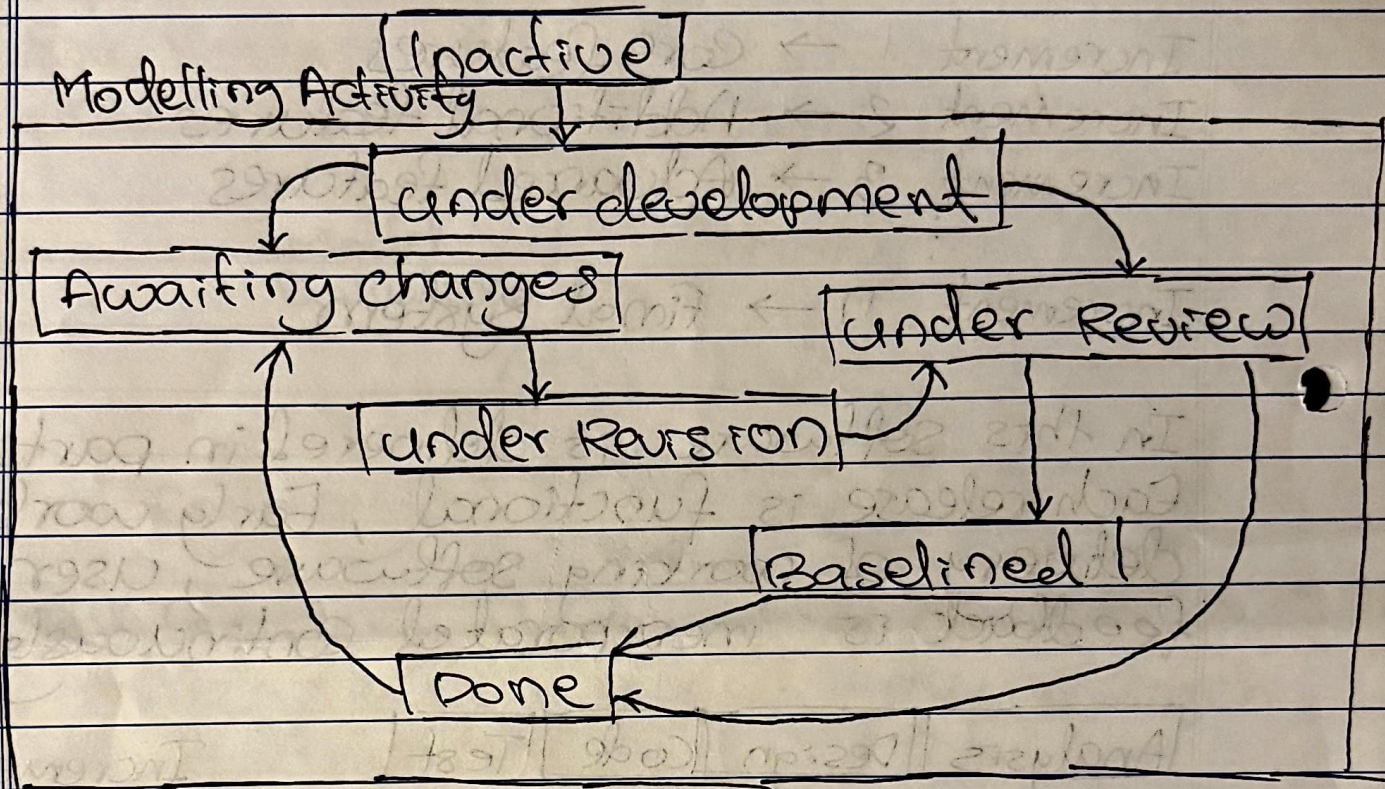


4. Concurrent Model

This model allows all software activities to run parallelly. Each activity is one of several states. As soon as something changes in one activity, the states of other activities also change.

This makes the model flexible and ideal

for project where requirements evolve continuously.
For eg.:



- ① Inactive: Phase hasn't started yet
- ② under development: work is being created
- ③ Under review: work is checked and feedback is given
- ④ Awaiting changes: issues found, phase pause for updates
- ⑤ Under revision: Team makes corrections or modifications
- ⑥ Baseline: Accepted and frozen temporarily
- ⑦ Done: may reopen if needed